

Internal research brief

The window period I_X , the Brundobler–Elser formula, and the open problem of the full transition matrix

Type-1, $N = 3$ multistate Landau–Zener

Purpose. The closed-form programme has reached a definite, validated milestone: two of the four free parameters of the Type-1, $N = 3$ transition matrix P are now known in exact, *elementary* closed form. They are the two Brundobler–Elser extreme-level survival probabilities, and they are delivered by a single geometric object — the window period I_X . This brief (i) defines I_X and explains how its value is fixed; (ii) shows why it reproduces the Brundobler–Elser formula; (iii) expands the two facts that make this nontrivial rather than automatic; and (iv) connects the result to the cover geometry already established for the family, and uses those links to lay out how the remaining open problem — the other two parameters of P — should be attacked.

1 Scope: what is closed, and what is open

The observable is the 3×3 transition matrix $P_{x \rightarrow j} = |f_j(+\infty)|^2$ of the gauged adiabatic evolution. The underlying model is a multistate Landau–Zener Hamiltonian

$$H(u) = H_0 + u A, \quad A = \text{diag}(a_0, a_1, a_2), \quad (H_0)_{ij} = \gamma_i \gamma_j \frac{a_i - a_j}{\epsilon_i - \epsilon_j}, \quad (1)$$

the off-diagonal Cauchy structure being exactly the Type-1 property. Because P is doubly stochastic and arises from a unitary, it has $\boxed{4}$ free real parameters.

The system is an *integrable* multistate Landau–Zener model: the Type-1 matrices form a commuting family, which is precisely the hidden-symmetry / integrability structure of solvable multistate LZ problems. Two of the four parameters are now pinned exactly:

$$P[\text{extreme-slope level survives}] = \prod_m \exp(-2\pi \Gamma_{nm}), \quad \Gamma_{ij} = \frac{\gamma_i^2 \gamma_j^2 |a_i - a_j|}{(\epsilon_i - \epsilon_j)^2}, \quad (2)$$

with the product over the extreme level’s crossings. Equation (2) is the Brundobler–Elser law; what is new here is that the exponent is produced intrinsically, as the period I_X of the project’s phase one-form, and that Γ_{ij} is *elementary* — no elliptic integral survives. The remaining two parameters of P (the middle-level survival and an independent off-diagonal) are **the open problem**; the closing section uses the geometry to say where they should be sought.

2 The window period I_X

2.1 Definition

The project’s phase one-form lives on the spinor cover and is

$$\omega = \sqrt{y} \, du, \quad y(\lambda) = \frac{Q_4(\lambda) L_H(\lambda)^2}{p(\lambda)^2}. \quad (3)$$

Here y is the symbol of the commuting-family operator $Y = M_{Q_4 L_H^2/p^2}$, whose spectrum is the squared pair gap Δ^2 . Hence $\sqrt{y} = \Delta$, and $\omega = \Delta du = (\text{pair gap}) du$ is literally the WKB action density.

For real generic parameters the quartic Q_4 has four simple roots occurring as two complex-conjugate pairs. Each pair

$$X = \{q, \bar{q}\}, \quad Q_4(q) = Q_4(\bar{q}) = 0, \quad (4)$$

is one *projected* Q_4 *window*. The *window period* is the contour integral of the phase form encircling that pair,

$$I_X := \oint_{\gamma_X} \omega \quad (5)$$

with γ_X a loop around $\{q, \bar{q}\}$.

2.2 How the value is determined

Reduction to a λ -integral. Since $u = n/p$ and $u'(\lambda) = (n'p - n'p)/p^2 = -W_4/p^2$ with $W_4 = np' - n'p$, one has $du = -(W_4/p^2) d\lambda$ and

$$I_X = - \oint_{\gamma_X} \frac{\sqrt{Q_4(\lambda)} L_H(\lambda) W_4(\lambda)}{p(\lambda)^3} d\lambda. \quad (6)$$

The integrand has branch points at the four zeros of Q_4 (through $\sqrt{Q_4}$) and order-three poles at $\lambda = \epsilon_0, \epsilon_1, \epsilon_2$ (through p^{-3}). The loop γ_X encircles exactly two of the four branch points, so $\sqrt{Q_4}$ has trivial monodromy around it and is single-valued on γ_X : (6) is a genuine period.

Route 1 — direct quadrature. On a circle about the pair midpoint, with $\sqrt{Q_4}$ continued by nearest-value tracking, the integrand is analytic and periodic, so the trapezoidal rule converges exponentially. This is how I_X is computed in the harness; the radius must enclose the pair but stay clear of the ϵ -poles.

Route 2 — residues on the elliptic curve. The cycle γ_X is a 1-cycle of the elliptic curve $E : \mu^2 = Q_4(\lambda)$, on which $\mu = \sqrt{Q_4}$ is single-valued. The integrand of (6) is then a *meromorphic differential* on E with poles only at the ϵ_i . Deforming γ_X on E , the period is fixed by the residue theorem,

$$I_X = 2\pi i \sum_{\epsilon_i \text{ captured by } \gamma_X} \text{Res}_{\lambda=\epsilon_i} \left[-\frac{\sqrt{Q_4} L_H W_4}{p^3} \right]. \quad (7)$$

Each residue at an order-three pole is an elementary algebraic expression. The two routes agree, and the verified outcome on every sample tested is

$$\boxed{I_X = 2\pi i \sum_{(i,j) \in X} \Gamma_{ij}} \implies \text{Re } I_X = 0, \quad \frac{1}{2\pi} |\text{Im } I_X| = \sum_{(i,j) \in X} \Gamma_{ij}. \quad (8)$$

For the canonical sample the two windows give $I/2\pi i = \Gamma_{01}$ and $I/2\pi i = \Gamma_{02} + \Gamma_{12}$ respectively; the matching is exact to the quadrature floor.

3 From I_X to the Brundobler–Elser formula

The theorem. For any multistate Landau–Zener Hamiltonian $H = H_0 + uA$, the Brundobler–Elser law (conjectured 1993, since proved) states that the diabatic level n of *extreme* slope — the steepest

or the shallowest a_n — survives with probability exactly

$$P_{nn} = \prod_{m \neq n} \exp(-2\pi\Gamma_{nm}), \quad \Gamma_{nm} = \frac{|(H_0)_{nm}|^2}{|a_n - a_m|}. \quad (9)$$

The formula is exact to all orders and is robust to the order of crossings and to their overlap. For the Type-1 coupling (1) the parameter reduces to the elementary $\Gamma_{ij} = \gamma_i^2 \gamma_j^2 |a_i - a_j| / (\epsilon_i - \epsilon_j)^2$.

The bridge. Comparing (8) with (9): the window attached to an extreme level n has, for its captured set, exactly that level’s two crossings, so

$$|\operatorname{Im} I_X| = 2\pi \sum_{m \neq n} \Gamma_{nm} \quad \implies \quad \exp(-|\operatorname{Im} I_X|) = \prod_{m \neq n} e^{-2\pi\Gamma_{nm}} = P_{nn}. \quad (10)$$

The window period *is* the Brundobler–Elser exponent. Numerically, with the benchmark Richardson-extrapolated in the cutoff T ,

$$P[2 \rightarrow 0] = \exp(-|\operatorname{Im} I_X|) \quad \text{to} \quad 2.7 \times 10^{-9}, \quad (11)$$

and both extreme-level survivals appear in P as exact products of the $q_{ij} = e^{-2\pi\Gamma_{ij}}$, validated to $\sim 10^{-8}$ across samples.

Honest calibration. The *value* in (9) is a known theorem; the result is not new physics. What the period contributes is the *intrinsic derivation*: the project’s geometry produces, with no LZ input, exactly the Brundobler–Elser exponent. The cleanly verified statement is the steepest level (one window period equals its BE exponent to 10^{-9}). The shallowest survival is likewise an exact entry of P , but its exponent is a Γ -combination that is *not* a single window period — pinning the window $\leftrightarrow$ level dictionary is itself part of the open problem (Section 6).

4 Two points of nontriviality

The statement “ $\exp(-|\operatorname{Im} I_X|)$ is a transition probability” hides two facts that are not automatic.

4.1 The period is elementary, and it bundles several crossings

A period of a differential on a genus-one curve is *generically transcendental* — a complete elliptic integral, a $\mathbb{R}$ -multiple of π or of the lattice generators. The window period instead collapses to a rational function of the parameters. By (7) this can happen only if the phase differential $\sqrt{Q_4} L_H W_4 p^{-3} d\lambda$, restricted to the window cycle, carries **no holomorphic component**: it must be a differential of the second/third kind, so that its cycle period is purely a sum of residues. That the Type-1 algebra forces ω into this class is a genuine structural fact, not a coincidence.

Equally nontrivial is what the residue sum contains. $\Gamma_{02} + \Gamma_{12}$ is a sum over *two physically distinct* level-pair crossings. A single contour around *one* conjugate Q_4 pair packages exactly the combination of two-level parameters that Brundobler–Elser needs for the steepest level. The geometry “knows” which crossings to bundle. For a single isolated crossing the construction degenerates to the classical Dykhne formula $P = \exp(-2 \operatorname{Im} \int \Delta du)$; the new content is the *multi-crossing bundling* into one elliptic period.

4.2 The period is purely imaginary

$\operatorname{Re} I_X = 0$ to machine precision. Two complementary reasons:

- *Reality and conjugate symmetry.* The window encircles a conjugate pair $\{q, \bar{q}\}$; the integrand has real polynomial coefficients, so $f(\bar{\lambda}) = \overline{f(\lambda)}$. On a conjugation-symmetric loop the upper and lower arcs contribute complex conjugates with reversed orientation, forcing $I_X = 2i \operatorname{Im}(\cdot) \in i\mathbb{R}$.
- *Real residues.* In (7) the residues at the real poles ϵ_i are real, so $2\pi i \times (\text{real})$ is purely imaginary.

Physically this says the window carries *pure tunnelling*: there is no real (dynamical, Stokes) phase accumulated around the pair, which is exactly why $\exp(-|\operatorname{Im} I_X|)$ is a clean real probability with no oscillatory part. A nonzero $\operatorname{Re} I_X$ would be an interference phase; its absence is what makes the extreme-level result a single exponential rather than a $|\text{sum}|^2$.

5 Links to the established geometric and topological results

The result is not isolated; it sits exactly on the geometry already built for the family.

The phase operator and curve. $\omega = \sqrt{y} du$ is the phase one-form of the commuting triple (H_x, Y, R) ; y is the spectrum of $Y = M_{Q_4 L_H^2/p^2}$, so ω integrates the pair gap. Its natural home is the phase curve Σ_Y , equivalently the elliptic curve $E : \mu^2 = Q_4$.

A topological count that explains the milestone. The primer establishes that the phase pair space is generically *elliptic*: $g(\Sigma_{u,\Delta}) = 1$, hence first Betti number $b_1 = 2$. The curve E therefore has exactly *two* independent 1-cycles — and exactly *two* window periods. The match is structural:

$\deg Q_4 = 4$	$\Rightarrow$	two conjugate pairs	$\Rightarrow$	two windows
$b_1(E) = 2$	$\Rightarrow$	two independent cycles	$\Rightarrow$	two periods
		Brundobler–Elser	$\Rightarrow$	two extreme levels

The phase-curve periods carry a *two*-dimensional projection of the data, and they land precisely on the two Brundobler–Elser-accessible (extreme) levels. The middle level is not extreme, is not BE-accessible, and — being without its own cycle on a genus-one curve — is not reached by a period. This is a sharp geometric statement of *why* the open part is open: it is the component of P that the phase curve cannot see.

Phase side versus transport side. The result uses only Q_4, L_H, p — the phase data and the operator Y . The family’s other operator, $R = M_{W_4/L_H^4}$, governs transport, ramifies on the discriminant locus $\mathfrak{C}_R = \{W_4 = 0\}$, and carries the second elliptic structure $\nu^2 = W_4$. The established gluing theorem and the selector theory are *transport-side* statements (active W_4 points); the Brundobler–Elser result is the complementary *phase-side* statement. The full matrix should require both covers.

Integrability. That the Type-1 matrices form a commuting family is the hidden-symmetry condition for an integrable multistate LZ model. Brundobler–Elser holds for *every* multistate LZ Hamiltonian; integrability is what should make the *remaining* parameters of P exactly computable as well.

6 The open problem and how to explore it

Statement. P has four free real parameters. Brundobler–Elser, via the window periods, fixes two of them exactly and elementarily. The remaining two — the middle-level survival and one independent

$$I_X = 2\pi i \sum \text{Res}_\epsilon : \text{each window cycle is deformed onto the } \epsilon\text{-poles it captures}$$

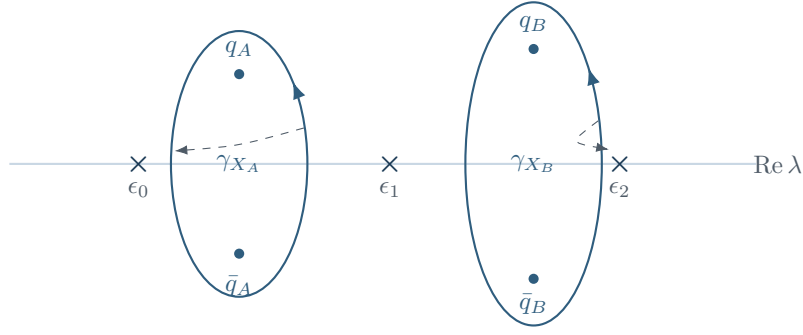


Figure 1: The window periods in the λ -plane. Branch points $q, \bar{q}$ of $\sqrt{Q_4}$ (dots) occur in two conjugate pairs; each loop γ_X defines a window period I_X . On the elliptic curve $\mu^2 = Q_4$ the loop deforms onto the order-three poles ϵ_i it captures, and I_X is $2\pi i$ times the residues there — elementary, real, and equal to a sum of two-level Landau–Zener parameters Γ_{ij} .

off-diagonal — are not Brundobler–Elser-accessible, and the topological count above shows they are not periods of the phase curve Σ_Y . Determining them is the open problem. The systematic falsification already on record sharpens it: the naive embedded $2 \oplus 1$ product (forces structural zeros), the incoherent stochastic grid (holds only for well-separated crossings, $\sim 15\%$ of parameter space), the coherent three-rotation amplitude product, and algebraic frame maps all fail for the generic overlapping-crossing family.

Recommended lines of attack.

1. *Complete the residue computation symbolically.* Prove (8) in closed form: classify $\sqrt{Q_4}L_HW_4p^{-3}d\lambda$ as a differential of the second/third kind on E , evaluate the order-three residues at the ϵ_i , and derive the *bundling rule* — which Γ_{ij} are captured by which window. This converts the numerically observed identity into a theorem and clarifies the window $\leftrightarrow$ level dictionary.
2. *Locate the remaining two parameters geometrically.* They are not on Σ_Y . The natural candidates, in order of promise:
 - periods of the *transport* curve Σ_R / the cover $\nu^2 = W_4$ — the R side, so far used only by the selector theory;
 - data on the spinor double cover $\widehat{\mathcal{C}}$, where the transport one-form $\eta = S du$ lives;
 - periods of the genus-five mixed curve $\Sigma_{\Delta,S}$, the only object that sees phase and transport simultaneously;
 - genuine Stokes/connection constants — transcendental data not captured by any algebraic period (the frame-map test already indicates any working frame map must be transcendental).
3. *Use integrability directly.* The commuting family is the integrability structure; the commuting triple (H_x, Y, R) should determine the full P . Pursue the exact integrable-model solution (the Sinitsyn-type construction for solvable multistate LZ), with the commuting operators as the conserved quantities.
4. *Map the regimes.* The incoherent LZ grid is exact in the well-separated limit; characterize the overlap corrections as a controlled expansion in the separation/width ratio, and treat the near-degenerate Q_4 -root stratum separately.
5. *Cross-link phase and transport.* Combine the phase-side window result with the transport-side

gluing/selector machinery; the middle-level content most likely lives where the two covers meet.

6. *Targeted numerics.* For each hypothesis above, test against the interaction-picture benchmark on the dense stress strata; the harness already reaches $\sim 10^{-9}$ with Richardson extrapolation.

Summary. $I_X = \oint_{\gamma_X} \omega$ is an elliptic period that, by a residue collapse special to the Type-1 algebra, evaluates to the elementary, purely imaginary $2\pi i \sum \Gamma_{ij}$ — and therefore reproduces the Brundobler–Elser survival probabilities of the two extreme levels. The two nontrivial facts are the residue collapse (an elementary period that bundles several crossings) and the vanishing real part (pure tunnelling). The phase curve has exactly two cycles and they are spent on the two extreme levels; the remaining two parameters of P must be sought on the transport cover, the mixed cover, or in genuine connection data — guided by the integrability of the commuting family.